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SECJ 2203 Software Engineering

Semester 01, 2024/2025

Section 01

PROJECT PROPOSAL

KADA SYSTEM

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1. Introduction

Explaining using NABC: the need (N), approach (A), benefit (B) and competitor (C) of the proposed system KADA, as follows:

Need (N)	The challenges faced by KADA align with common issues in similar agencies where manual or outdated systems lead to bottlenecks, data inaccuracies, and slower response times. [3] Process management research and user input from other agencies highlight the necessity of efficient workflows and user-friendly technology to satisfy an agency's changing needs. The proposed system solution might allay these worries by providing a dependable, scalable and effective system that is suited to KADA's unique operational requirements and allows for improved member record and financial application management. KADA would benefit from these enhancements.

Approach (A)	The proposed solution is a cloud-based platform with integrated AI, designed to automate and streamline KADA's internal processes, particularly member registration and loan applications. This system features a centralized **member management database** for efficient registration, updates, and data access, allowing KADA to manage member information with ease. The **loan application process** is further enhanced by AI-powered automation that reduces manual tasks by evaluating loan eligibility, processing applications, and providing real-time status updates for applicants. To support decision-making, a **data analytics dashboard** monitors agency performance, tracks loan statuses, and identifies areas where process improvements could be implemented. Additionally, a **mobile app interface** provides members with a user-friendly platform to apply for loans, check the status of their applications, and receive notifications regarding approvals or requests for additional information, making the entire experience more accessible and efficient for all users.
Benefit (B)	The proposed XYZ system brings important benefits to KADA and its stakeholders by improving usability, performance, and cost-efficiency. With a **user-friendly mobile app**, members can easily register, apply for loans, and get real-time updates and notifications, which reduces the need for in-person visits and paperwork, making the process simple and accessible. The system also **boosts efficiency and performance** by automating tasks like loan eligibility checks and member data entry, minimizing human errors, speeding up approvals, and allowing KADA to manage more applications without increasing staff. [4] Additionally, a **data analytics dashboard** gives KADA valuable insights into

	loan distributions, member demographics, and any bottlenecks in the process, helping leaders make better decisions, use resources wisely, and continuously improve operations. The **cloud-based design** cuts down on physical infrastructure and maintenance costs, while AI automation reduces labor costs, leading to significant savings over time. Altogether, these benefits create a smoother experience for members, increase productivity for KADA staff, and lower costs, supporting the agency's sustainable growth.

Competitor (C)	The KADA system stands out from other agency management platforms by focusing specifically on agricultural needs and incorporating advanced AI automation tailored to KADA's processes. **Flexibility and customization** set KADA apart: while general platforms like Salesforce [5] and SAP [6] offer broad functions, they lack the specific tools needed by agricultural agencies. KADA, however, is designed with dedicated features for managing member registration and loan processing, aligning closely with KADA's workflow. **AI-driven automation** is another unique strength of KADA. Unlike competitors that often use basic, rule-based automation, KADA's AI tools handle complex tasks like eligibility assessments, application processing, and real-time status updates, learning from past data to become more accurate over time. This advanced automation reduces processing times and errors, especially during busy periods, giving KADA a significant advantage in efficiency.

2. Existing Systems

The existing system operates manually, which limits its efficiency and productivity. Nevertheless, it does offer several main services and features, as listed below:

1. Membership Registration

This is a structured process for KADA's staffs to become part of their members, granting them accessible to a variety of cooperative benefits and services.

The current member registration system relies on a manual process, necessitating individuals to complete information on printed forms. During registration, various personal details, such as name, identification number, address, and others are gathered. Afterward, KADA staff will need to review the registration documents again. This method is not only time-consuming but also susceptible to errors and places a significant strain on human resources.

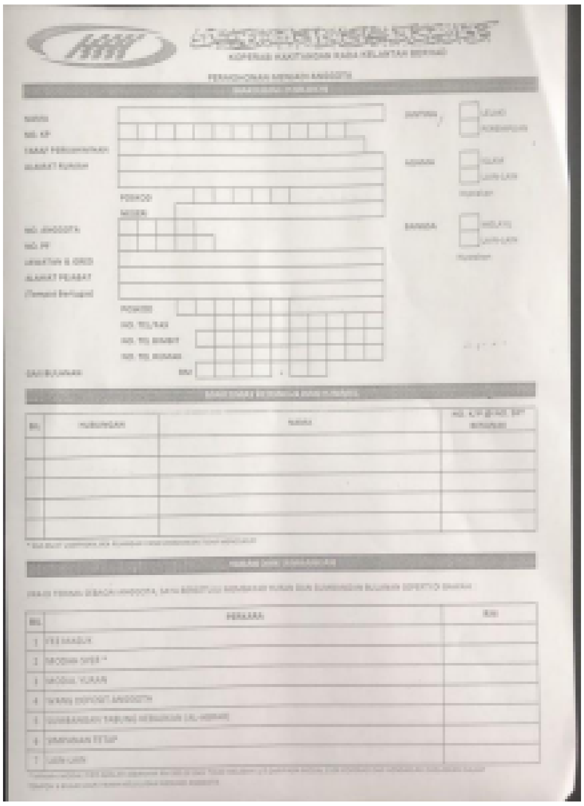


Figure 3.1 printed Member Registration Form

2. Loan Applications

Their loan application seeks financial assistance for personal, educational, or business needs. The goal of the loan application process is to be clear and efficient, allowing cooperative members to obtain the required funds quickly. Currently, the loan application system is manual too, necessitating that information be entered on printed forms. In

addition, the qualified members are not able to check on their past transactions and records that is hard for them to look back for their total loans and payments if they lost their statements.

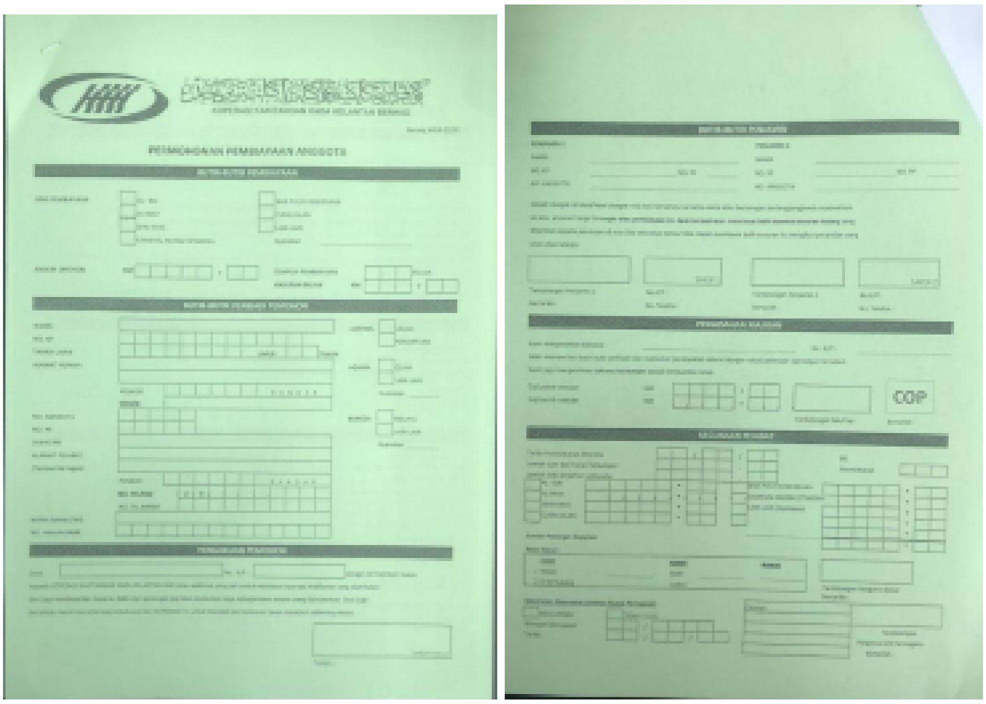


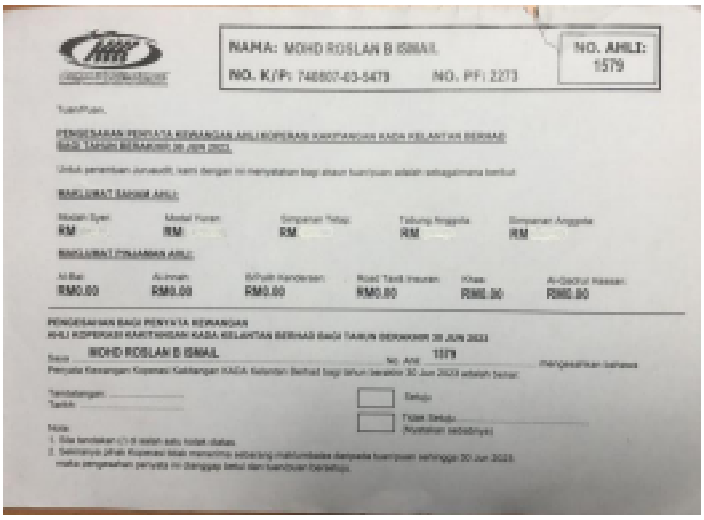
Figure 3.2 Member Loan Applications Form

3. Member Financial Statements

Koperasi Kakitangan KADA Kelantan Berhad currently uses a manual system for generating and distributing member financial statements. This involves collecting financial data manually from various records and compiling it into individual member statements. These statements include details such as the member's name, IC number, shares held, loans taken, and savings.

Once the statements are prepared, they are reviewed by staff to ensure accuracy. After verification, the statements are handed out to members. Members then sign and return the verification portion to confirm receipt and accuracy.

This manual process is time-consuming, labor-intensive, and prone to errors due to manual data entry. It also places a significant burden on the cooperative's limited human resources and delays members' access to their financial information. Implementing an automated system would streamline this process, reduce staff workload, and provide members with timely and accurate financial information.



NAMA: MOHD ROSLAN B ISMAIL
NO. K/P: 740807-03-5479 NO. PF: 2273
NO. AHLI: 1579

Figure 3.3 Member Financial Statement

<u>Problem with Existing System</u>

The registration system at KADA company is currently handled manually, showing signs of low efficiency and productivity, as described below. Even though it offers only a limited number of essential services and features, there is room for improvement. Recognizing the constraints of the existing system emphasizes the necessity for a complete overhaul to enhance the system. Moving towards a new computerized system could greatly elevate their services and features. This new system will bring about automation, streamlined processes, an improved website for better performance, and an expanded array of services for users.

1. Insufficient registration process

   In the current system, all registration procedures are manually processed, like customers filling out paper forms to register and staff handling each form individually. Personal information, both private and sensitive, is predominantly stored in hard copies during registration. This method not only poses security risks but also makes it challenging to retrieve or locate information when necessary. Additionally, the registration papers require further collection, rearrangement and scrutiny by staff at the end. It really takes much more time than doing the registration online.

2. Inconvenient Status Checking

   In the existing process, members are unable to check the status directly, they must either call or wait for an email from the company to obtain the current status, whether it is about the registration approval or payment status. This method is not that convenient, as staff have to manually send emails instead of having an automated system on their website that segregates the status. Improvements are necessary to modernize this outdated system to enhance the efficiency of the processes.

3. Heavy Dependence on Human Resources

The existing manual system faces an issue as it heavily depends on human resources. All the steps starting from the registration and application for the loan are done manually. If the number of human resources decreases beyond a certain point, it will impact productivity and efficiency in the workplace. To address the issue of heavy dependence on human resources, the implementation of automation solutions is crucial as it can help streamline processes, minimise errors, and free up employees to focus on more strategic tasks that require human intervention.

4. Inefficient and human error

We know that retrieving and managing physical documents is time-consuming and the manual system heavily depends on human resources. This might lead to the chances of human error rising when staff forget to email the current status, missing payment receipts, or failing to gather customer information and also missing information. These inefficiencies will cause delays in service delivery and mistakes in data processing, affecting overall customer satisfaction and also might disrupt the entire development process as incomplete or inaccurate data may result. Therefore, the company requires a more well-designed system.

5. Security Concerns of Manually Stored Customer Forms and Information

In the digital age, safeguarding customer information is paramount for maintaining trust and ensuring compliance with data protection regulations. The cooperative currently relies on manual storage of customer forms and information in physical cupboards. This method poses significant security risks that need urgent attention to protect sensitive data and uphold our commitment to customer privacy. As we can see, employees, cleaning staff, and visitors may easily access the cupboards, potentially leading to data breaches. Furthermore, it is challenging to track who accessed or modified the physical documents, making it difficult to audit and ensure the accountability of data handling practices.

Features	**Existing System**	**Proposed System**
Membership registration	Yes	Yes
Loan application	Yes	Yes
Application status checking	No	Yes
Check financial status and statements	Yes	Yes
Check monthly and annually report	No	Yes
Check profit and loss from account	No	Yes

3. Proposed System

The proposed KADA Registration System is a fully integrated digital platform platform aimed at enhancing the efficiency, security, and overall user experience of Koperasi Kakitangan KADA

Kelantan Berhad. It replaces the current manual processes, which are time-consuming, prone to errors, and heavily reliant on human resources.

The system allows new users to register online by submitting personal details such as name, address, and identification data. Upon submission, This data is processed automatically, verified for accuracy, and stored securely in a digital database, and provides real-time feedback on the status of the application, whether it is successful, pending, or rejected. Once a user becomes a registered member, they gain access to several convenient services, such as checking the real-time status of their registration, loan applications, and payments through an online portal. Members can also manage their savings, view financial statements, and track potential dividends from their investments, receiving automated email or SMS notifications regarding their financial activities.

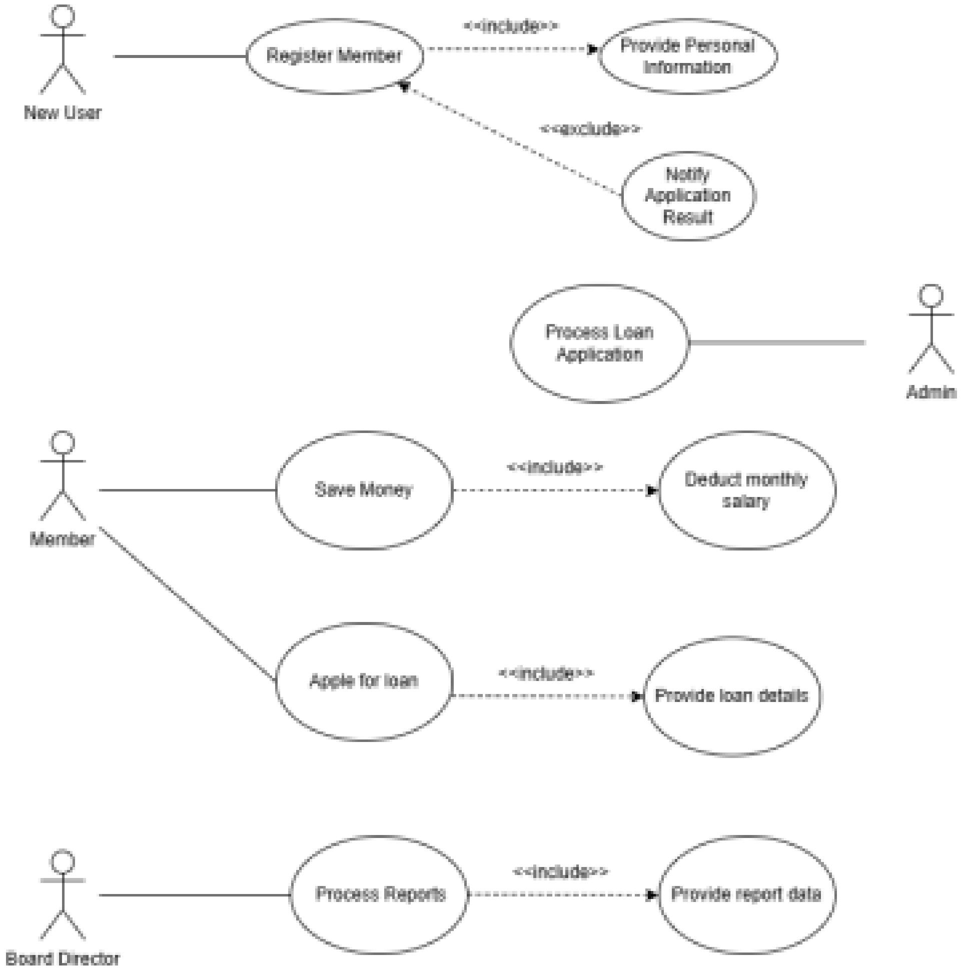
For the administrative staff (KADA employees), the system reduces the burden of manual data handling and processing. Admin are responsible for reviewing member applications, verifying submitted information, and authorizing loan applications based on pre-set criteria. For the admin, the system simplifies tasks by providing a centralized platform for reviewing member applications and loan requests. Admins can verify application details, approve or reject submissions based on eligibility criteria, and manage the digital records securely, reducing the need for paper-based processes. They can also update the status of applications and ensure that members are promptly notified of any changes. The system minimizes human error by automating many of the routine processes, freeing up staff to focus on more strategic and high-priority tasks.

Board Directors, who oversee high-level decisions, are involved in reviewing and approving larger or more critical loan applications. Their role is to ensure that these applications align with the cooperative's strategic goals and regulatory requirements. They have the authority to make final decisions on both member and loan applications, with their approval or rejection logged into the system for transparency and future reference.

Overall, the KADA Registration System brings about a more efficient, secure, and user-friendly environment for all stakeholders. It reduces the burden on human resources, improves data

management, enhances communication, and ensures that sensitive member information is stored securely. Also, it enhance security measures through features like Two-Factor Authentication (2FA), and providing a seamless experience for users. By automating key functions and providing a streamlined user interface, the system aims to elevate the cooperative's service delivery and operational capabilities, fostering greater member satisfaction and organizational productivity.

Use Case Diagram for Proposed System



4. Software Process Model

For the software process model, we selected the waterfall model for the KADA system. This is a linear, sequential life cycle model. It is easy to learn and utilize. In principle, each phase must be completed before proceeding to the next, and no phases overlap.

Planning Phase (Duration: 14 days)

The first stage is the planning phase. During this phase, our team gathered all of the detailed requirements from our stakeholder, KADA, via online meeting. These thorough criteria involve defining the project scope and gathering information about the existing system's problems through an interview with a KADA representative. After gathering all the information, we documented it, drafted an initial version of the proposal, and submitted it to our lecturer, Dr. Iqbal, for review.

Design Phase (Duration: 9 days)

The second stage is the design phase. During this phase, we constructed logical and data flow diagrams (DFD), such as the context diagram, diagram 0, and child diagram. In addition, we produced the physical DFD, which focused on the context diagram, diagram 0, child diagram, CRUD matrix, partitioning, event response table, structure chart, and system architecture diagram. Next, we'll use Figma to design a system wireframe. We then demonstrated our system wireframe to Dr. Iqbal, who checked it and made some suggestions. After we had completed all of the suggestions, we modified our prototype and submitted the final version of the report.

Testing Phase (Duration: 9 days)

The testing phase is the third stage. We are testing and demonstrating the Figma prototype to our lecturer, who will then provide feedback for our improvement.

5. Project Schedule

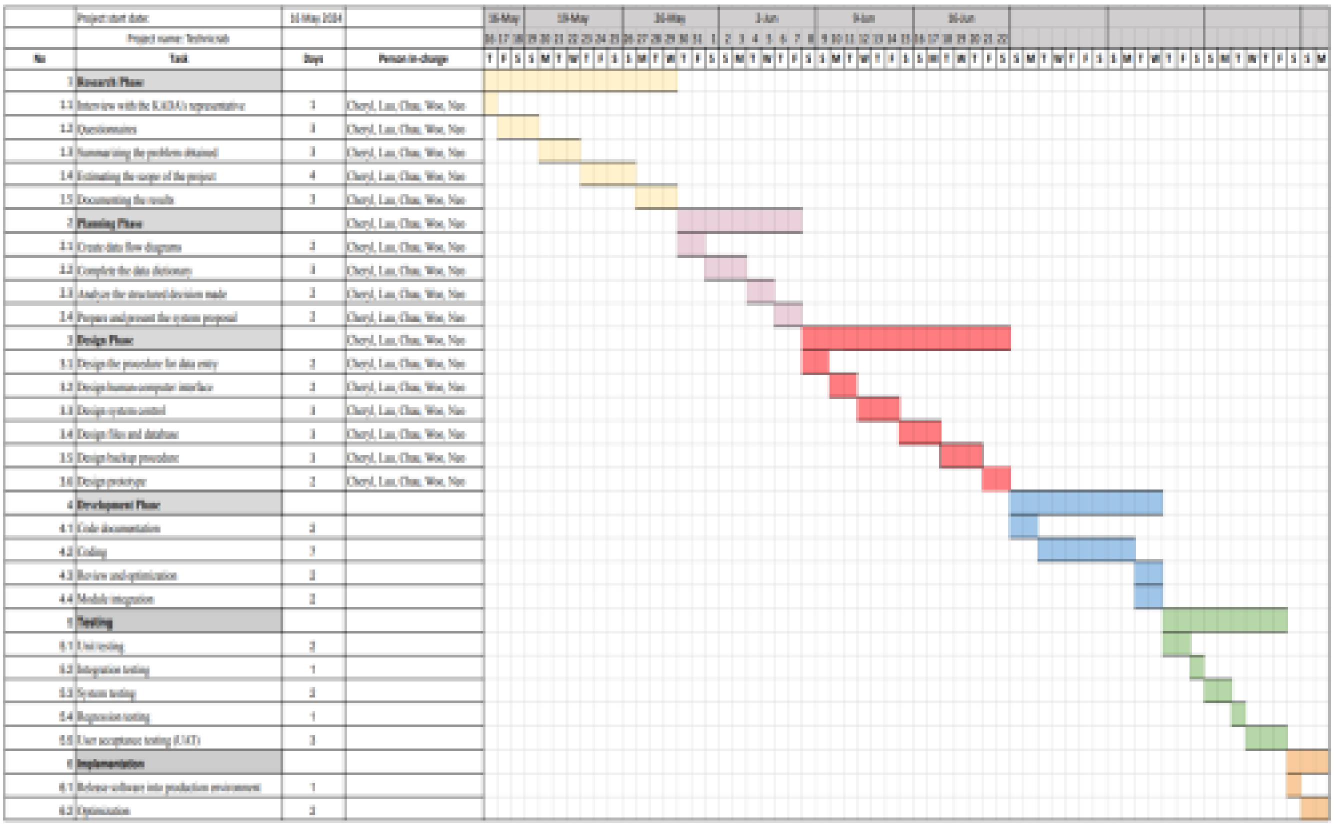


Figure 2: Gantt Chart

6. References

[1] Sommerville, I. (2016). “Software Engineering”, 10th Edition, US: Pearson.

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[6] SAP. "SAP Software Solutions | Business Applications and Technology." *SAP*, [Online]. Available: https://www.sap.com/sea/index.html. Accessed: Oct. 25, 2024.

[7] Dennis, A., Wixom, B. H., & Roth, R. M. (2008). *Systems analysis and design*. John wiley & sons.

Appendices

Example: Screenshot

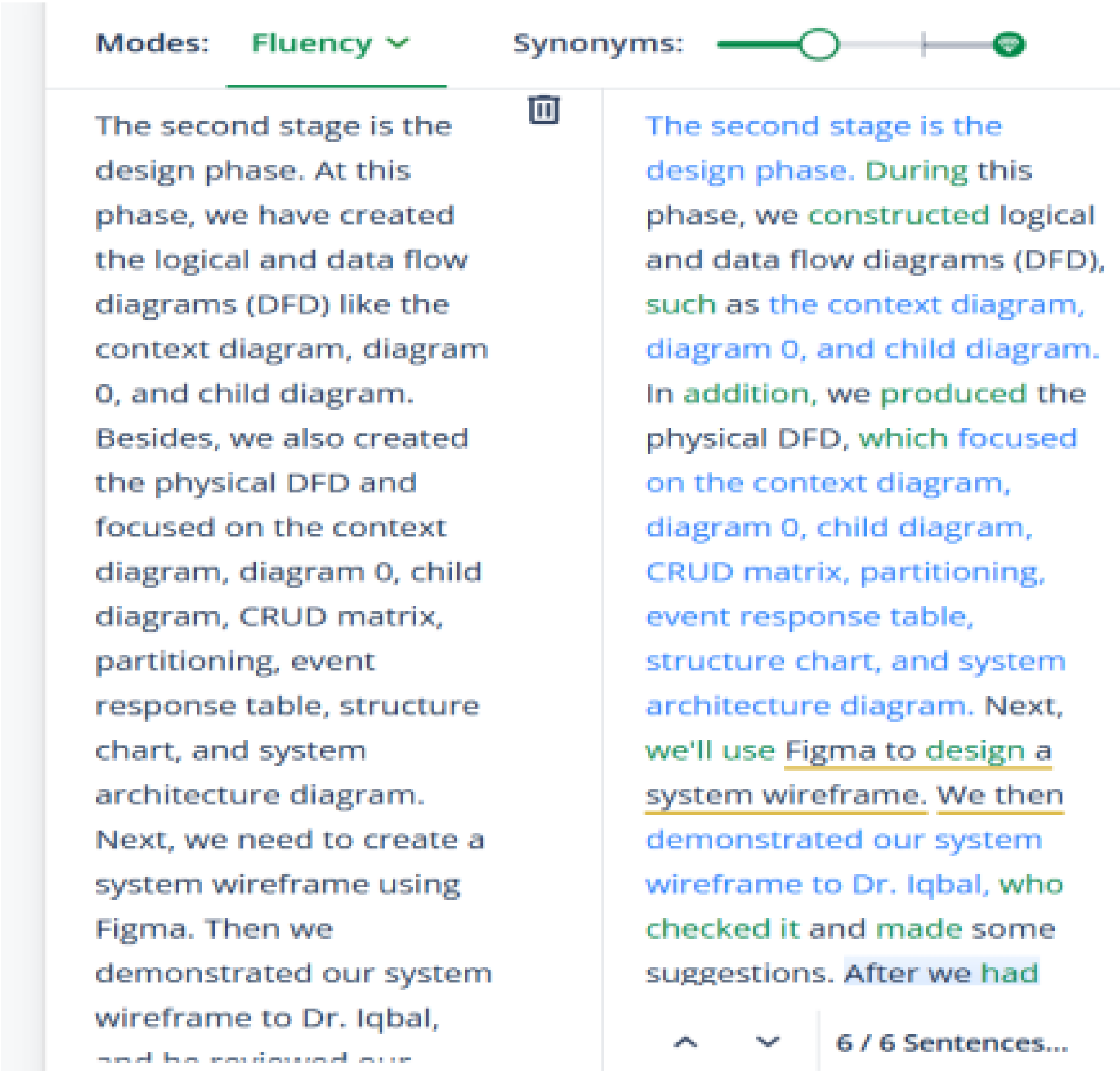

Figure: Screenshot Paraphase

Example: URL

https://quillbot.com/paraphrasing-tool